

1.1. What is the purpose of this course?

The purpose of this course is to provide you with a solid understanding of the fundamental concepts and techniques of the field. This includes a deep understanding of the underlying principles, as well as the ability to apply these principles to solve complex problems. The course is designed to be both challenging and rewarding, and we hope that you will find it to be a valuable experience. The course is divided into several modules, each of which covers a different aspect of the field. The first module covers the basics of the field, while the subsequent modules cover more advanced topics. The course is designed to be self-paced, so you can learn at your own speed. We will provide you with all the resources you need to succeed, including lecture notes, assignments, and a final exam. We hope that you will find this course to be a valuable experience, and we look forward to seeing you in class.

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Die lineare Induktion Methode

Bevor wir die lineare Induktion anwenden, überlegen wir uns, was eine lineare Induktion ist. Eine lineare Induktion ist eine Methode, um zu zeigen, dass eine Aussage für alle natürlichen Zahlen n gilt. Die lineare Induktion besteht aus zwei Schritten: 1. Basisfall: Man zeigt, dass die Aussage für $n=1$ gilt. 2. Induktionsschritt: Man zeigt, dass wenn die Aussage für $n=k$ gilt, dann gilt sie auch für $n=k+1$. Wenn man diese beiden Schritte gezeigt hat, dann gilt die Aussage für alle natürlichen Zahlen n .

Beispiel: Zeige, dass die Summe der ersten n natürlichen Zahlen $1+2+\dots+n$ gleich $\frac{n(n+1)}{2}$ ist.
 Basisfall: Für $n=1$ gilt $1 = \frac{1(1+1)}{2} = 1$.
 Induktionsschritt: Angenommen, die Aussage gilt für $n=k$. Dann gilt $1+2+\dots+k = \frac{k(k+1)}{2}$. Wir zeigen nun, dass die Aussage auch für $n=k+1$ gilt.

$$1+2+\dots+k+(k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}$$

Das ist genau die Aussage, die wir zeigen wollten. Also gilt die Aussage für $n=k+1$. Da die Aussage für $n=1$ gilt und der Induktionsschritt bewiesen ist, gilt die Aussage für alle natürlichen Zahlen n .

Beispiel 2: Zeige, dass die Summe der ersten n ungeraden Zahlen $1+3+\dots+(2n-1)$ gleich n^2 ist.

$$1+3+\dots+(2n-1) = n^2$$

Basisfall: Für $n=1$ gilt $1 = 1^2$.
 Induktionsschritt: Angenommen, die Aussage gilt für $n=k$. Dann gilt $1+3+\dots+(2k-1) = k^2$. Wir zeigen nun, dass die Aussage auch für $n=k+1$ gilt.

$$1+3+\dots+(2k-1)+(2k+1) = k^2 + (2k+1) = (k+1)^2$$

Das ist genau die Aussage, die wir zeigen wollten. Also gilt die Aussage für $n=k+1$. Da die Aussage für $n=1$ gilt und der Induktionsschritt bewiesen ist, gilt die Aussage für alle natürlichen Zahlen n .

Beispiel 3: Zeige, dass die Summe der ersten n Quadratzahlen $1^2+2^2+\dots+n^2$ gleich $\frac{n(n+1)(2n+1)}{6}$ ist.
 Basisfall: Für $n=1$ gilt $1^2 = \frac{1(1+1)(2\cdot 1+1)}{6} = 1$.
 Induktionsschritt: Angenommen, die Aussage gilt für $n=k$. Dann gilt $1^2+2^2+\dots+k^2 = \frac{k(k+1)(2k+1)}{6}$. Wir zeigen nun, dass die Aussage auch für $n=k+1$ gilt.

$$1^2+2^2+\dots+k^2+(k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2 = \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} = \frac{(k+1)(k(2k+1) + 6(k+1))}{6} = \frac{(k+1)(2k^2+7k+6)}{6} = \frac{(k+1)(k+2)(k+3)}{6}$$